

Harun Emrecan Karabağ

Unity Game Developer | C# Software Engineer | Gameplay & Systems

Ankara, Türkiye | +90 507 049 3574 | harunemrecankarabag@gmail.com

Portfolio: hekyetmisbes.github.io | Itch.io: hekyetmisbes.itch.io | GitHub: github.com/hekyetmisbes | LinkedIn: linkedin.com/in/harunemrecankarabag

Professional Summary

Computer Engineering graduate and current M.Sc. student in Computer Animation and Game Technologies, focused on Unity, C#, gameplay systems, software architecture, mobile development, UI/HUD implementation, simulation, and interactive product development. Hands-on experience spans casual and mobile games, narrative puzzle design, driving simulation, backend software engineering, research-oriented simulation, data persistence, third-party services, testing, and performance optimization.

I have developed projects from early concept and technical architecture through implementation, debugging, validation, iteration, and published playable builds. My engineering approach combines clean and reusable code, data-driven design, algorithmic problem solving, player-facing polish, technical documentation, and performance-conscious development. I am comfortable taking ownership in small multidisciplinary teams while continuing to deepen my skills in scalable game systems and software engineering.

Professional Focus

- End-to-end Unity/C# development: gameplay features, progression, persistence, UI/HUD, input, third-party services, testing, optimization, and build preparation.
- Systems-oriented problem solving: data-driven architecture, state management, event-driven flows, algorithms, AI/pathfinding, validation, and performance-aware runtime design.
- Product and player experience: rapid prototyping, readable interaction loops, responsive feedback, mobile-friendly UI thinking, game jam production, and iterative polishing.
- Software engineering discipline: OOP, SOLID fundamentals, modular code, Git workflows, backend development, documentation, debugging, testing, and maintainable project organization.

Core Skills

Game Development: Unity 6, Unity 2D/3D, C#, mobile and casual games, simulation prototypes, narrative games, platformers, puzzle/quiz mechanics, game jam prototyping

Gameplay & Systems: player movement, interactions, mission flow, scoring, progression/economy foundations, achievements, skill trees, daily challenges, collectibles, win/loss logic, game-state flow

Unity UI/UX Integration: Unity UI, Canvas, menus, HUD elements, score/result/settings screens, responsive UI thinking, Figma-to-Unity workflow, player feedback systems

Architecture & Code Quality: OOP, SOLID fundamentals, modular architecture, composition root, ScriptableObjects, state machines, event-driven systems, interfaces, serialization/migration, Repository Pattern, Dependency Injection, CQRS fundamentals

Algorithms & AI: A* pathfinding, graph traversal, heuristic cost design, spatial-grid indexing, Dictionary, HashSet, Queue, Stack, priority queues, agent-based simulation, optimization, forecasting, AI-supported modeling

Testing & Debugging: Unity Test Framework, EditMode/PlayMode test foundations, xUnit, validation, bug reproduction, debugging, refactoring, regression-testing concepts, QA checklist mindset

Performance: object pooling, spatial partitioning, distance-based update throttling, performance-aware update loops, allocation-aware architecture, profiling concepts, CPU/GPU fundamentals

Services, Data & Backend: Firebase Realtime Database, UnityWebRequest, SQLite, JSON persistence, ASP.NET Core 8, Entity Framework Core, FluentValidation, data modeling, logging

Tools & Workflow: Git, GitHub, GitLab, Visual Studio, Docker fundamentals, Figma, technical documentation, build testing, iterative development

Languages: Turkish - Native; English - B2+ written and spoken

Selected Game & Software Projects

Honey Tycoon - Unity 6 Casual / Idle Systems Foundation

PC/mobile-targeted idle management project built around data-driven production, upgrades, research, automation, economy, and long-term progression systems.

Technologies: Unity 6, C#, URP, ScriptableObjects, Unity Test Framework, JSON, Git

- Designed a scalable technical foundation with versioned GameState, Profile, Economy, Inventory, and World models to keep progression data explicit and maintainable.
- Implemented JSON serialization and migration foundations, a bootstrap composition root, ScriptableObject-based definitions, and editor-side validators for configuration and state integrity.
- Separated runtime, editor, and test assemblies with clear system boundaries and added EditMode/PlayMode test foundations for configuration, bootstrap behavior, and core state workflows.
- Documented architectural and performance rules for atomic economy transitions, frame-rate-independent production, event-driven updates, save/version compatibility, mobile performance budgets, and maintainable feature growth.

Delivery Driver - Open-City Delivery Simulation Prototype

[GitHub](#) | [Playable Build](#)

Stylized Unity 3D driving and delivery project combining player control, mission/progression systems, route guidance, minimap navigation, NPC traffic, persistence, UI/HUD feedback, and runtime optimization.

Technologies: Unity 3D, C#, Input System, Cinemachine, A*, spatial indexing, SQLite, EasyRoads3D, NPC traffic AI, object pooling, Git

- Built player-facing systems including delivery missions, pickup/drop-off flow, money and XP rewards, levels, achievements, skill-tree progression, daily challenges, cargo damage, statistics, settings, and save/load flows.
- Implemented incremental A* road-graph pathfinding using Dictionary/HashSet search state, queued priority handling, heuristic costs, waypoint graphs, and spatial-grid indexing for efficient candidate lookup.
- Developed route/minimap guidance and HUD feedback including navigation markers, speedometer-style information, mission state communication, and progression feedback.
- Built simulation and optimization systems around NPC traffic, obstacle avoidance, pooled spawning, spatial partitioning, distance-based update throttling, and simplified far-AI behavior.
- Published a downloadable Windows build on Itch.io and maintained the project through iterative updates, bug fixes, gameplay improvements, and development logs.

Fog Bridge - Atmospheric Narrative Puzzle Game

[Playable Build](#)

GameDev.tv Game Jam 2026 project about two isolated guards gradually building understanding through paper-plane messages across a fog-covered chasm.

Technologies: Unity, C#, narrative puzzle design, fictional alphabet decoding, interaction flow, resource gathering, bridge rebuilding progression, Itch.io release

- Designed and released a compact atmospheric narrative experience centered on distance, fear, language, and connection.
- Built a readable interaction loop where the player receives paper-plane messages, decodes a fictional alphabet, replies across the fog, gathers materials, and gradually rebuilds a forgotten bridge.
- Focused on quiet emotional pacing, symbolic worldbuilding, storybook-inspired presentation, and interaction-driven progression that transforms isolation into understanding.
- Completed the project within a game-jam production cycle and published a downloadable Windows build, demonstrating rapid scope control, implementation, polishing, and release discipline.

Flag Quiz - Android Mobile Quiz Game

[GitHub](#)

Mobile quiz game where players identify country flags through a simple, accessible, score-driven loop with remote data integration and mobile-focused UI.

Technologies: Unity 2D, C#, Android, Firebase Realtime Database, UnityWebRequest, Unity UI, Figma, Git

- Built the mobile-friendly quiz loop with randomized flag presentation, randomized answer options, answer validation, score tracking, round progression, success/failure states, audio feedback, and high-score persistence.
- Integrated Firebase Realtime Database for remote question data and used UnityWebRequest to retrieve flag images, strengthening practical third-party service and remote-content integration skills.
- Designed UI concepts in Figma and iterated Android-focused interactions around clarity, readability, responsive player input, and short-session usability.

The Platform - 2D Platformer Game

[GitHub](#) | [Playable Build](#)

Released cyberpunk-themed 2D platformer built around movement, hazards, collectibles, level progression, and straightforward completion goals.

Technologies: Unity 2D, C#, platformer mechanics, level flow, collectibles, ShaderLab/HLSL, Git

- Developed 2D movement, obstacle interaction, level progression, collectible-driven objectives, and completion conditions for a complete playable loop.
- Designed levels around collecting stars, navigating hazards, and completing objectives with clear visual and gameplay feedback.
- Explored ShaderLab/HLSL alongside core Unity development and published a downloadable Windows build with install instructions and packaged executable.

Movidle Game - Movie Guessing Game

[GitHub](#)

Data-backed guessing game based on IMDb Top 250 information, comparing player guesses through attributes such as genre, year, director, country, and star.

Technologies: Java, game logic, validation systems, CSV/data handling, UI flow

- Developed the core guessing logic where the system randomly selects a target movie and compares player predictions across multiple movie attributes.
- Implemented limited-attempt gameplay, feedback-driven comparisons, success/failure states, and CSV-backed movie selection.
- Used the project as an early foundation for game-state logic, validation, data-driven behavior, and player feedback before later expanding into Unity/C# development.

Tri-Ink Track - Unity 2D Systems Prototype

Additional Unity 2D project focused on event-driven state flow and reusable runtime systems.

Technologies: Unity 2D, C#, event-driven state flow, Queue/HashSet-based object pooling

- Worked with event-driven state transitions and Queue/HashSet-based object pooling to practice reusable runtime system design and efficient object lifecycle management.

The Final Loop - Unreal Engine Interaction Prototype

Additional interactive project used to explore interaction-state gameplay outside the Unity ecosystem.

Technologies: Unreal Engine 5, Blueprints, interaction-state gameplay

- Explored Unreal Engine 5 and Blueprint-based interaction logic, broadening practical familiarity with alternative real-time development workflows and visual scripting.

Professional Experience

Co-Founder

Dec 2025 - Apr 2026

Fiktion Technology Studios

Ankara, Türkiye

- Contributed to early-stage game and software initiatives through technical direction, feature planning, prioritization, documentation, implementation ownership, and coordination in a small multidisciplinary team.

Software Engineering Intern

Jul 2025 - Aug 2025

Erciyes University Research Dean's Office / IT

Kayseri, Türkiye

- Developed maintainable ASP.NET Core 8 and Entity Framework Core components with data modeling, validation, debugging, logging, documentation, and iterative improvements.
- Applied Clean Architecture, CQRS, Repository Pattern, Dependency Injection, FluentValidation, Docker fundamentals, xUnit tests, and Git-based collaborative development practices.

Hardware Engineering Intern

Jul 2024 - Aug 2024

Fotonik Teknoloji A.Ş.

Kayseri, Türkiye

- Supported hardware system design and improvement, hands-on testing, system analysis, issue investigation, and technical problem solving in a cross-disciplinary engineering environment.

Research Intern - 2247-C STAR Fellow

Feb 2024 - Jul 2024

TÜBİTAK

Türkiye | Advisor: Asst. Prof. Saltuk Buğra Selçuklu

- Worked on agent-based simulation, optimization, forecasting, and AI-supported modeling, strengthening algorithmic thinking, experimentation, quantitative reasoning, and structured problem solving.

Education

Hacettepe University

2026 - Present

M.Sc. in Computer Animation and Game Technologies | GPA: 3.81 / 4.00 | Expected Graduation: Jan 2027

Erciyes University

2020 - 2025

B.Sc. in Computer Engineering | GPA: 3.00 / 4.00

Certificates

- Introduction to Game Design - Epic Games / Coursera - Nov 2025
- Unity 401 - Turkcell Geleceği Yazarlar - Jul 2024
- More C# Programming and Unity - University of Colorado Boulder / Coursera - Mar 2024 | [Certificate](#)
- C# Class Development - University of Colorado Boulder / Coursera - Mar 2024
- Introduction to C# Programming and Unity - Coursera | [Certificate](#)
- Version Control: Git and GitHub - BTK Akademi - Oct 2023 | [Certificate](#)
- Hypercasual Fundamentals - BTK Akademi - Jan 2023 | [Certificate](#)